



OFFICE OF THE CHAIRPERSON BOARD OF STUDIES IN PHYSICS,
UNIVERSITY COLLEGE OF SCIENCE, OSMANIA UNIVERSITY,
HYDERABAD

NO 09/CB067/2024

DATE 2/8/2024

Engineering Physics Lab Syllabus for Osmania University Affiliated
Engineering Colleges
B.E (SEM-I&II) AICTE
Academic Year 2024-2025 Onwards (Common to All Branches)

Academic Year 2024-2025 Onwards (Common to All Branches)							
CourseCode: BS251PH	CourseTitle: Physics Lab					Core/Elective : Core	
Prerequisite	ContactHoursper Week				CIE	SEE	Credits
	L	T	D	P			
	-	-	-	3	25	50	1.5

Course Objective

- Master experimental procedures and programming techniques.
- Conduct experiments independently with precision and measurement accuracy
- Analyze data graphically and derive conclusions from graphs
- Evaluate experiment results critically and draw meaningful conclusions.
- Improve communication skills through group work and effective laboratory report writing.

Course Outcomes

- Apply the various procedures and programming techniques for the experiments.
- Demonstrate the experiment with task and take the measurement independently
- Examine the graphical representation data and estimate results from the graph.
- Compare and evaluate the results of the experiment and draw relevant conclusions
- Develop communication skills through working in groups in performing the laboratory Experiments and by writing laboratory reports.

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2/8/2024

HEAD

Department of Physics
University College of Science
Osmania University
Hyderabad- 500 007, TS

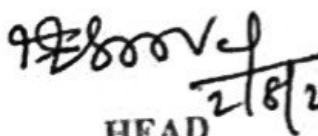
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List of Experiments

1. Determination of Dielectric Constant and Phase Transition Temperature of Dielectric Materials
2. To study the I-V Characteristics of P-N Junction Diode and Resistance Evaluation
3. Find the Electrical Conductivity and Energy Gap of Germanium (Ge) Crystal
4. Study Hall Effect in Semiconductors & find Hall Coefficients, Hall Voltage, and Conductivity
5. Study Characteristics of Thermistor: Determination of Constants A and B
6. Draw Hysteresis Loop for Ferromagnetic Material (B-H Curve)
7. Study V-I Characteristics of Solar Cell: Fill Factor and Series Resistance Calculation
8. Visualization Energy Levels of 1-Dimensional Potential Box Using Schrödinger Wave Equation in Python
9. Visualization of Allowed Energy Levels and Kronig-Penney Model in Python/MATLAB
10. Determine the Density and Elastic Properties of Oxide Glasses/polymers Using Machine learning algorithms
11. Calculate the Numerical Aperture (NA) and Acceptance Angle of Optical Fiber
12. Find the Wavelength of Laser Source using diffraction grating
13. To study the Transition Temperature Measurement of High-Temperature Superconductor
14. Find the Rigidity Modulus of Wire Using Torsional Pendulum
15. To estimate the pressure using optical fiber sensor.

Note: Minimum Eight experiments should be conducted in the semester


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